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Website



ACCEL-JS2000

NVIDIA IGX platform for Medical AI imaging application



FEATURE

- NVIDIA IGX Orin Industrial Module with 250 TOPS AI Performance
- Support RTX 6000 Ada GPU card with up to 1700 TOPS AI Performance
- ConnectX-7 High Speed Edge Networking 2 x 100Gbe
- Dedicated MCU to reassure the functional safety
- Built-in BMC controller to provide a standard-based interface to system management and recovery
- System Power design with 700W Medical PSU
- Excellent thermal management to maximize the performance and lower the noise
- 4K SDI capture card integration (Optional)
- Support optional 7" front screen
- Long Life cycle Support

DESCRIPTION

ACCEL-JS2000 is a compact yet powerful medical-grade AI box PC designed for edge computing in clinical environments. With NVIDIA® Jetson AGX Orin™, it enables real-time inference for diagnostic imaging, patient monitoring, and medical robotics.

- Powered by NVIDIA® Jetson AGX Orin™ with up to 250 TOPS AI performance
- Fanless and compact design for easy integration into medical devices
- Medical-grade safety certification (EN 60601-1 compliant)
- Rich I/O interfaces: HDMI, LAN, USB, CAN bus, and more
- Supports AI frameworks including TensorRT, CUDA, and DeepStream
- Optimized for 24/7 operation in clinical and surgical settings

APPLICATION

- Medical Imaging AI: Real-time enhancement, segmentation, and classification
- Patient Monitoring Systems: Edge analytics for vitals, ECG, and telemetry data
- Surgical Assistance: Supports AI-based navigation and robotic coordination
- Diagnostic Devices: Compact design for integration into ultrasound, endoscopy, etc.
- Telemedicine & Edge Analytics: On-device inference reduces latency and improves responsiveness
- Hospital AI Infrastructure: Deployable in nurse stations, ICUs, and ORs for intelligent data flow

SPECIFICATION

Main Specifications	
AI Engine	NVIDIA IGX Orin Soc Industrial Module
CPU	12-core Arm® Cortex®-A78AE v8.2
System Memory	64GB 256-bit LPDDR5 204.8 GB/s
Graphics	2,048-core NVIDIA Ampere architecture with 64 Tensor Cores
Storage	M.2 NVMe SSD x 1 2.5" SATA SSD x2 64GB eMMC 5.1
OS support	IGX-SW 1.1
Front Screen (Optional)	7" LED Panel with P-Cap Touch 400nits 1280 x 800
Safety	Infineon Aurix TC397
NVIDIA BMC	Aspeed AST2600 Microchip ERoT
I/O	
USB	4 x USB 3.2 Gen 2 Type A 1 x USB 3.2 Gen 2 Type C
Ethernet	2 x 100GbE QSFP28 ports 2 x 1GbE RJ45
Video Out	1 x Display Port 1.4a
Video input (Optional)	1 x 12G SDI +1 x HDMI 2.0 HDMI 2.0(Max resolution up to 4096x2160@60fps) 12G SDI(Max resolution up to 4096x2160@60fps)
Audio	3 x 3.5mm AU Jacks(MIC, Line-in and Speaker out)
Serial Port	2 x COM Port
Function Port	Grounding pin x 1
Front I/O	1 x Power Button with Power LED indicator 2 x USB 2.0 Type A
Wireless	1 x M.2 Key E Key for Wireless Module 1 x M.2 Key B for 5G/LTE module
Expansion I/O	1 x PCIe Gen5 Double Width Slot (x16) for A6000 1 x PCIe Gen5 Single Width Slot (x8)

SPECIFICATION

Mechanical and Environmental

Power 100V to 240V AC Input
700W Medical Power Supply

Operating Temperature 0°C ~ 35°C(32°F ~ 95°F)

Storage Temperature -20°C ~ 60°C(-4°F ~ 140°F)

Dimension 400 x 166 x 407 mm

Package Size 560 x 305 x 628 mm

Gross Weight 14 kg

Net Weight 12 kg

Certifications CE: EN 60601-1-2:2015(V4.0)
EN 60601-1:2006/A1:2013/A12:2014 (V3.1)
FCC: Part 18 Class B, Part 15 Class B
UL: ANSI/AAMI ES60601-1:2012 (V3.1)
cUL: CAN/CSA-C22.2 No. 60601-1:2014 (V3.1)